



NOVEMBER 21

Vocab Learning App

Team
Mo_Li

Contributor
Yijia Li, Yuanman Mu



Progress

01 Redesigning the design

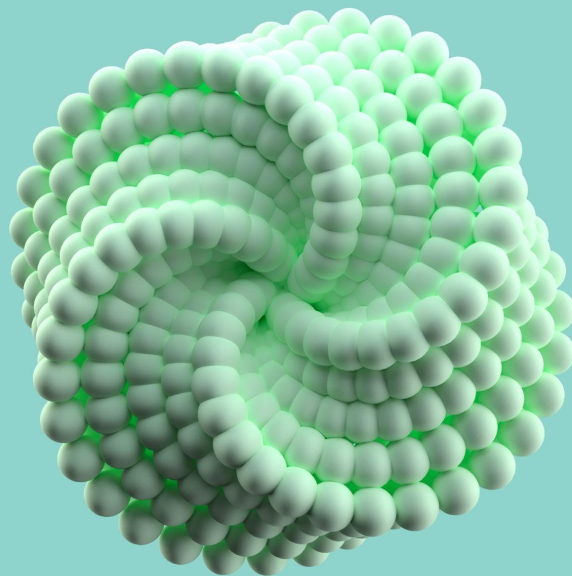
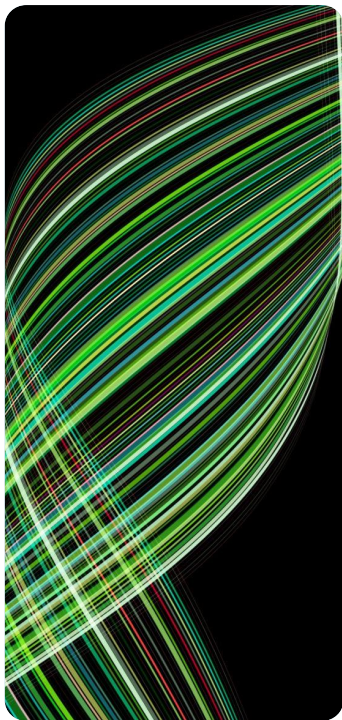
02 Modeling Databases

03 Handling technical issues



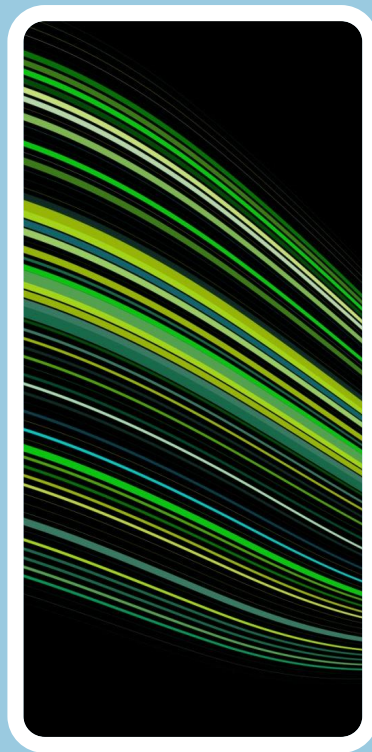
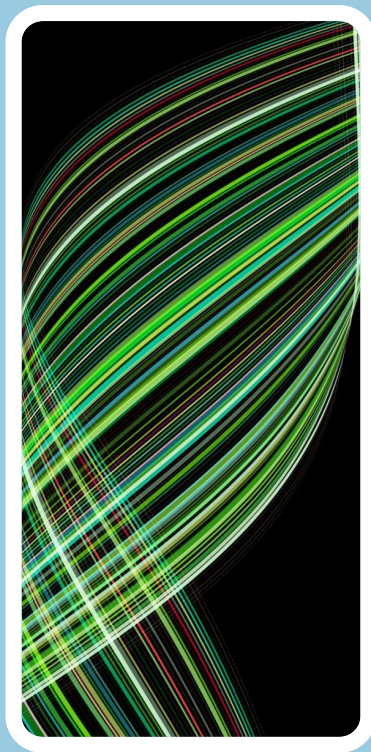
UI Redesign

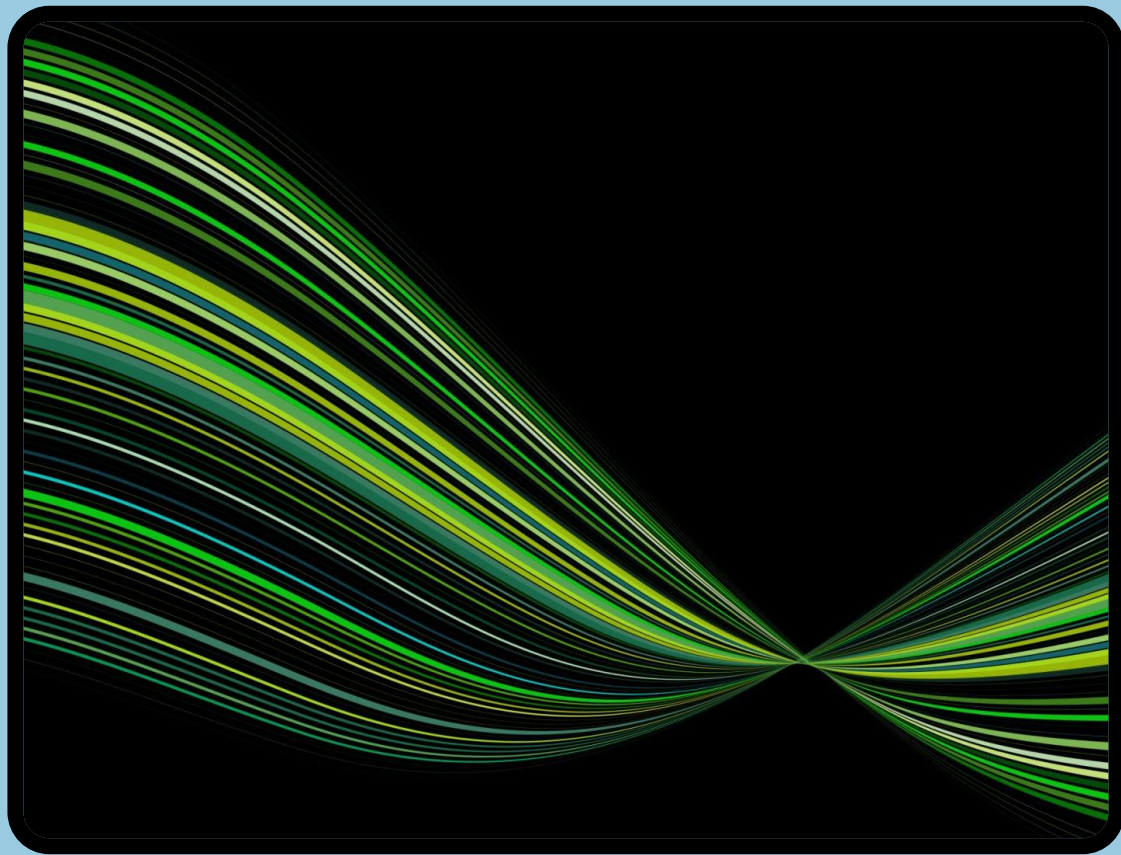




Company Name

Roadmap







Modeling Databases



Our Models

Vocabulary Table

Stores the main vocabulary data.

Quiz Records Tables

Stores information about quizzes and user performance.

Word Progress Tracking Table

tracking quiz performance and overall progress for each word.

Notification table

Tracks words that need to be sent as reminders in push notifications.

Workflow Based on Database Schema

1. Daily Word Selection

1. User specifies the number of words to learn for the day.
2. System queries the **Word Progress Tracking Table** to fetch words where:
 - status = 'unseen'.
3. Update these words' status to 'learning' and return them to the user.

2. Learning Progress

1. As the user interacts with each word: Update the status:
 - status = 'learning' for ongoing practice.
 - status = 'mastered' if the word is marked as mastered.

3. Quiz Triggers

1. Monitor the total number of words with status = 'learning'.
2. When the count reaches a multiple of 50, trigger a quiz session.
3. Include the following words in the quiz:
 - Words with status = 'learning'.

4. Quiz Execution:

1. Record the user's response in the **Word Progress Tracking Table**:
 - If the user answers correctly:
 - Update status = 'mastered'.
 - If the user answers incorrectly:
 - Keep status = 'learning'.

5. Notifications

1. Identify words that are still status = 'learning'.
2. Send reminders for these words based on their progress:
 - E.g., prioritize words with higher quiz_attempts or longer time since last update.



Handling technical issues

On-Device Sensors

- **Accelerometer:** Detect device orientation (portrait/landscape) for adaptive UI rendering.
- **Touchscreen Gestures:** Enable swipe-based navigation in flashcard mode.
- **Haptic Feedback:** Provide tactile responses for interactive elements (e.g., quiz answers).

1. AI is not reliable
2. English learning app for Chinese student(English/Chinese)
3. Where to find the list of vocabulary